Module 2: Answers

This section provides comprehensive answers to all questions (Q.3 and Q.4) from Module 2 of the Managerial Economics for Robotics (BRI501) examination.

Q.3 (a): Explain the role of economies of scale in robotics. (10 Marks)

Economies of Scale refer to the cost advantages that an enterprise obtains due to an increase in the scale of output (i.e., size of production). As production volume increases, the cost per unit of output decreases.

Role of Economies of Scale in Robotics

Economies of scale play a crucial and multi-faceted role in the robotics industry, affecting both the **manufacture of robots** and the **application of robots** in other industries.

1. In the Manufacture of Robots (Supply Side):

- **Fixed Cost Spreading:** The design, research, and development (R&D) of a new robot model (e.g., a collaborative robot or an industrial arm) involve high upfront fixed costs. As the number of units produced increases, this significant R&D cost is distributed over more units, drastically lowering the average cost per robot.
- **Specialization and Mass Production:** Large-scale production allows for highly specialized assembly lines, customized machinery for component manufacturing (e.g., precision gear cutting), and bulk purchasing of standardized parts (e.g., microprocessors, sensors, motors). This leads to greater efficiency and lower input costs.
- **Modular Design Benefits:** Many robots are built using modular components (e.g., standard controllers, actuators). Manufacturing these modules in extremely high volumes achieves large-scale economies, making the final robot cheaper to assemble.

2. In the Application of Robots (Demand Side):

- **Justifying High Initial Investment:** The initial capital investment required to deploy a robotic system (the robot itself, integration, safety systems) can be substantial. Economies of scale help justify this investment. In large-volume manufacturing (e.g., automotive), running the robot 24/7 (3-shift operation) spreads the high initial cost over millions of manufactured products, making the automation cost-effective.
- **Scale of Deployment:** Industries that deploy robotics across multiple sites or in very large facilities (e.g., massive e-commerce fulfillment centers) benefit from **purchasing economies**. They can negotiate deep discounts for buying hundreds or thousands of identical robotic units, further lowering the effective price per unit of automation.
- **Learning and Expertise:** Large-scale robotic deployment facilitates **learning economies**. As a company deploys more robots, its maintenance staff and operators gain experience, leading to faster troubleshooting, better system uptime, and reduced operational downtime and costs.

In summary, economies of scale are vital for making robotics an economically viable solution. They drive down the cost of robots, accelerate their adoption across various sectors, and ultimately enable industries to achieve greater efficiency in their own production processes.

Q.3 (b): Net Present Worth (NPW) Calculation and Proposal Selection (10 Marks)

The Net Present Worth (NPW) method is a capital budgeting technique that calculates the present value of all cash inflows and subtracts the present value of all cash outflows (initial investment). The acceptance criterion is: Select the proposal with the highest positive NPW.

The initial investment for both proposals is the cash outflow at Year 0: $P_0 = \text{Rs } -10,000$. The rate of interest is $i = 18\%$ or 0.18. The Present Worth Factor (PWF) for a given year (t) is $\frac{1}{(1+i)^t}$.

Proposal A: Calculation of Net Present Worth (NPW_A)

End of Year (t)	Net Cash Flow (Rs) (C_t)	PWF at 18% $\frac{1}{(1+0.18)^t}$	Present Worth (Rs) ($C_t \times PWF$)
0	$-10,000$	1.0000	$-10,000.00$
1	$3,000$	$\frac{1}{(1.18)^1} = 0.8475$	$3,000 \times 0.8475 = 2,542.37$
2	$7,000$	$\frac{1}{(1.18)^2} = 0.7182$	$7,000 \times 0.7182 = 5,027.78$
3	$6,000$	$\frac{1}{(1.18)^3} = 0.6086$	$6,000 \times 0.6086 = 3,651.81$
4	$3,000$	$\frac{1}{(1.18)^4} = 0.5158$	$3,000 \times 0.5158 = 1,547.45$
Total			$NPW_A = 2,769.41$

$$NPW_A = -10,000 + 2,542.37 + 5,027.78 + 3,651.81 + 1,547.45 = \mathbf{\text{Rs } 2,769.41}$$

Proposal B: Calculation of Net Present Worth (NPW_B)

End of Year (t)	Net Cash Flow (Rs) (C_t)	PWF at 18% $\frac{1}{(1+0.18)^t}$	Present Worth (Rs) ($C_t \times PWF$)
0	$-10,000$	1.0000	$-10,000.00$
1	$6,000$	0.8475	$6,000 \times 0.8475 = 5,084.75$
2	$6,000$	0.7182	$6,000 \times 0.7182 = 4,309.52$
3	$3,000$	0.6086	$3,000 \times 0.6086 = 1,825.90$
4	$3,000$	0.5158	$3,000 \times 0.5158 = 1,547.45$
Total			$NPW_B = 2,767.62$

$$NPW_B = -10,000 + 5,084.75 + 4,309.52 + 1,825.90 + 1,547.45 = \mathbf{\text{Rs } 2,767.62}$$

Comparison and Selection

Proposal	Net Present Worth (NPW)
A	Rs 2,769.41
B	Rs 2,767.62

Both proposals have a positive Net Present Worth, meaning both are economically viable. Since the NPW method dictates selecting the option with the highest present worth, **Proposal A should be selected** as its NPW (Rs 2,769.41) is slightly higher than Proposal B's NPW (Rs 2,767.62).

Q.4 (a): Explain in detail the various elements of cost with suitable examples. (10 Marks)

Cost is a fundamental concept in managerial economics, representing the expenditure incurred on resources to produce a specific good or service. Costs are categorized in various ways for decision-making.

1. Classification by Traceability (Direct and Indirect Costs)

Cost Element	Definition	Example in a Robotics Company
Direct Cost	Costs that can be directly and wholly attributed to a specific product, service, or cost center.	**Direct Material:** The steel, aluminum, sensors, and actuators that become part of the final robot arm. **Direct Labour:** The wages paid to the assembly line technicians who physically put the robot together.
Indirect Cost	Costs that cannot be directly traced to a specific product and are typically incurred for the benefit of multiple products or the entire organization. (Also known as Overheads).	**Factory Overheads:** Rent of the factory building, electricity for the entire plant, depreciation of general-purpose machinery. **Administrative Overheads:** Salaries of the CEO, accounting staff, and office supplies.

2. Classification by Behavior (Fixed, Variable, and Semi-variable Costs)

Cost Element	Definition	Example in a Robotics Company
Fixed Cost (FC)	Costs that remain constant in total, irrespective of the level of production or sales volume within a relevant range.	Annual lease payment for the R&D lab, annual software license fee for CAD/CAM tools, basic salaries of permanent managers.
Variable Cost (VC)	Costs that vary directly and proportionally with the level of production or sales volume.	Cost of materials (e.g., servo motors, wiring harnesses) used per robot, commission paid to sales agents per robot sold, packaging cost per unit.
Semi-Variable Cost	Costs that have both a fixed and a variable component.	Utility bills (a fixed monthly charge + a variable charge based on consumption), maintenance costs (a fixed retainer fee + variable costs for specific repairs).

3. Other Important Cost Concepts

Cost Element	Definition	Example in a Robotics Company
Opportunity Cost	The value of the next best alternative that is foregone when a choice is made.	If a company invests Rs 10 million in a new automated warehouse system, the opportunity cost is the expected return they could have earned by investing that Rs 10 million in a new marketing campaign or treasury bonds.
Sunk Cost	A cost that has already been incurred and cannot be recovered. It is irrelevant for future decision-making.	The Rs 5 million spent two years ago on R&D for a failed robot prototype. This cost cannot be recovered and should not influence the decision to launch a *new* robot project today.
Marginal Cost	The additional cost incurred by producing one more unit of a good or service.	If producing the 100th robotic arm costs Rs 2,00,000, then the marginal cost of the 101st arm is the extra cost incurred to produce it, which might be lower due to bulk material purchase.

Q.4 (b): Rate of Return (ARR) Calculation (10 Marks)

The Rate of Return (RoR), often referring to the **Accounting Rate of Return (ARR)**, is calculated by dividing the average annual net profit by the initial investment (or average investment).

The formula used here is based on the initial investment:

$$\text{Rate of Return (ARR)} = \frac{\text{Average Annual Net Profit}}{\text{Initial Investment}} \times 100$$

Given Data:

- Initial Investment = Rs $20,00,000$
- Annual Net Profit = Rs $3,50,000$
- Life of the Project = 10 years
- Salvage Value = Nil

Calculation:

In this case, since the annual net profit is constant and given directly, it also serves as the average annual net profit.

$$\text{ARR} = \frac{\text{Rs } 3,50,000}{\text{Rs } 20,00,000} \times 100$$

$$\text{ARR} = 0.175 \times 100$$

$$\text{ARR} = \mathbf{17.5\%}$$

Conclusion:

The Rate of Return (Accounting Rate of Return) for the new business line is 17.5%. This means the